

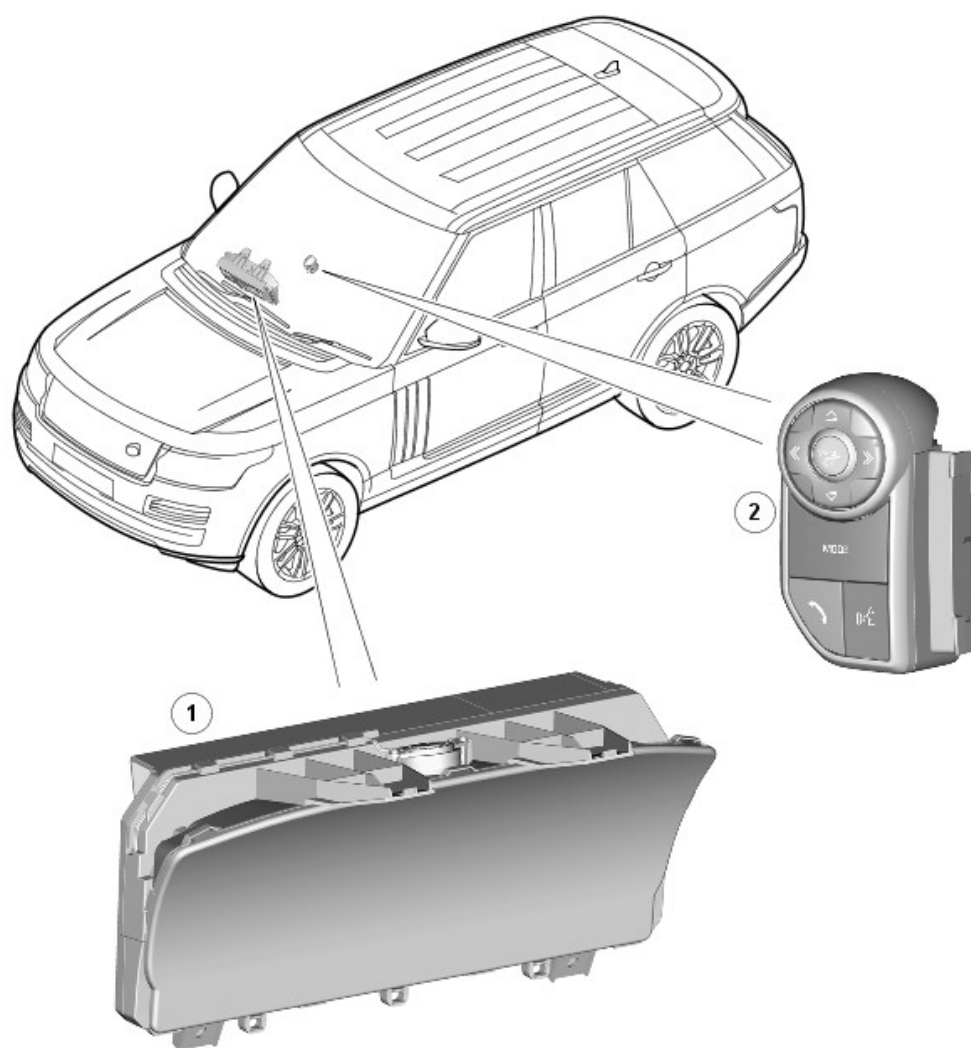
2016.0 RANGE ROVER (LG), 413-01

INSTRUMENT CLUSTER

DESCRIPTION AND OPERATION

COMPONENT LOCATION

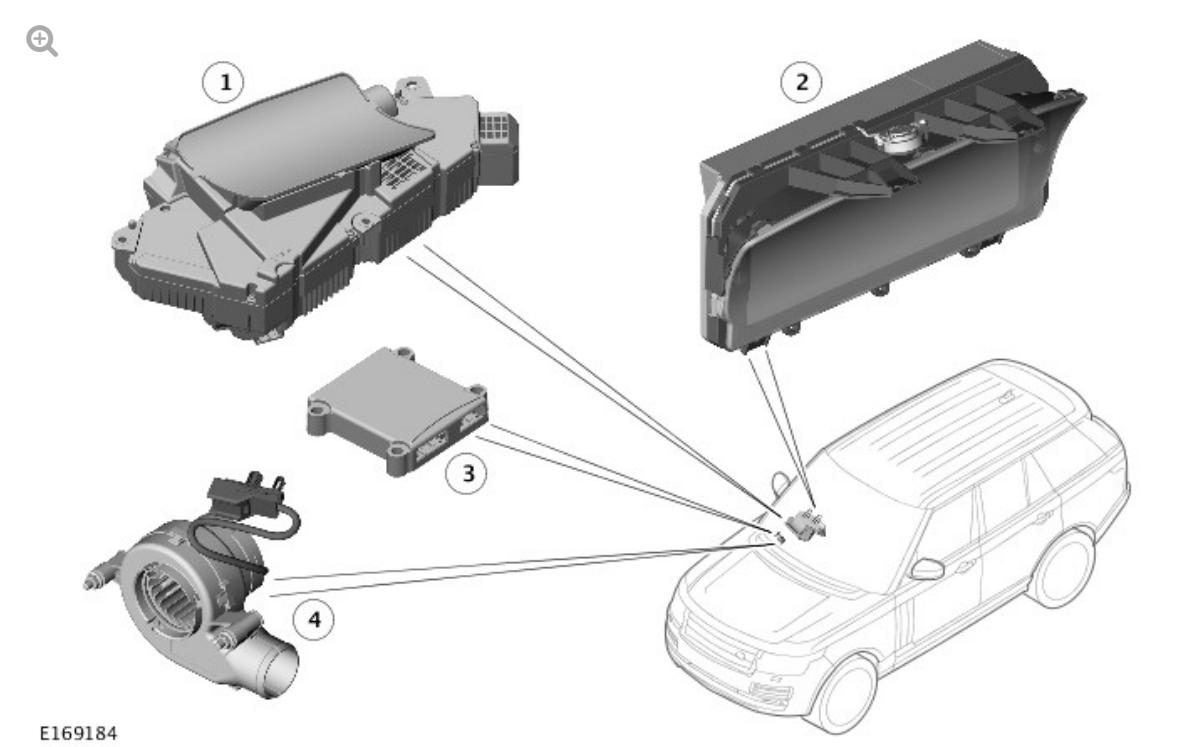
INSTRUMENT CLUSTER



E149325

ITEM	DESCRIPTION
1	Instrument Cluster (IC)
2	Left Steering Wheel Switchpack (Menu Control)

HUD (HEAD UP DISPLAY)



ITEM	DESCRIPTION
1	Instrument Cluster (IC)
2	Head Up Display (HUD)
3	HUD Cooling Fan Control Module (HUDCFCM)
4	HUD Cooling Fan Motor

OVERVIEW

INSTRUMENT CLUSTER

The Instrument Cluster (IC) is a high definition Thin Film Transistor (TFT) display.

The display is continually reconfigured to prioritize and refine the information presented to the driver. All displays are 'virtual' with the speedometer and the tachometer being the dominant features of the display. Two hardwired warning indicators are in the cluster; the security warning indicator and a primary airbag/Supplemental Restraint System (SRS) warning indicator. These warning indicators are Light Emitting Diodes (LED's). All other warning indicators are generated within the IC TFT screen.

Two ambient light sensors; one on either side of the instrument cluster, (to ensure the brightest lighting conditions are detected) are used to adjust cluster illumination in response to prevailing lighting conditions.

The IC features a number of warning indicators. The warning indicators illuminate in one of four colors which indicate the level of importance of the warning as follows:

- Red = Warning.
- Amber = Caution.
- Green = System operative.
- Blue = Headlamp high beam operative.

A feature and driver preference menu is available to allow the driver to select certain features and functions of the IC and change them to their personal preference. A menu control 'joy pad' is located on the left steering wheel switchpack and allows selection of the displayed functions and navigation of the menus. When selected, the menu is displayed in the TFT screen which allows access to a number of vehicle functions.

HEAD UP DISPLAY (HUD)

The HUD is a transparent display that presents data without the driver requiring to look away from their viewpoint.

A virtual image is displayed on the windscreen which appears at a distance of approximately 2 meters, giving the impression that the image appears around the end of the bonnet.

This function is designed to provide an increase in safety and convenience by maximising eyes on the road time.

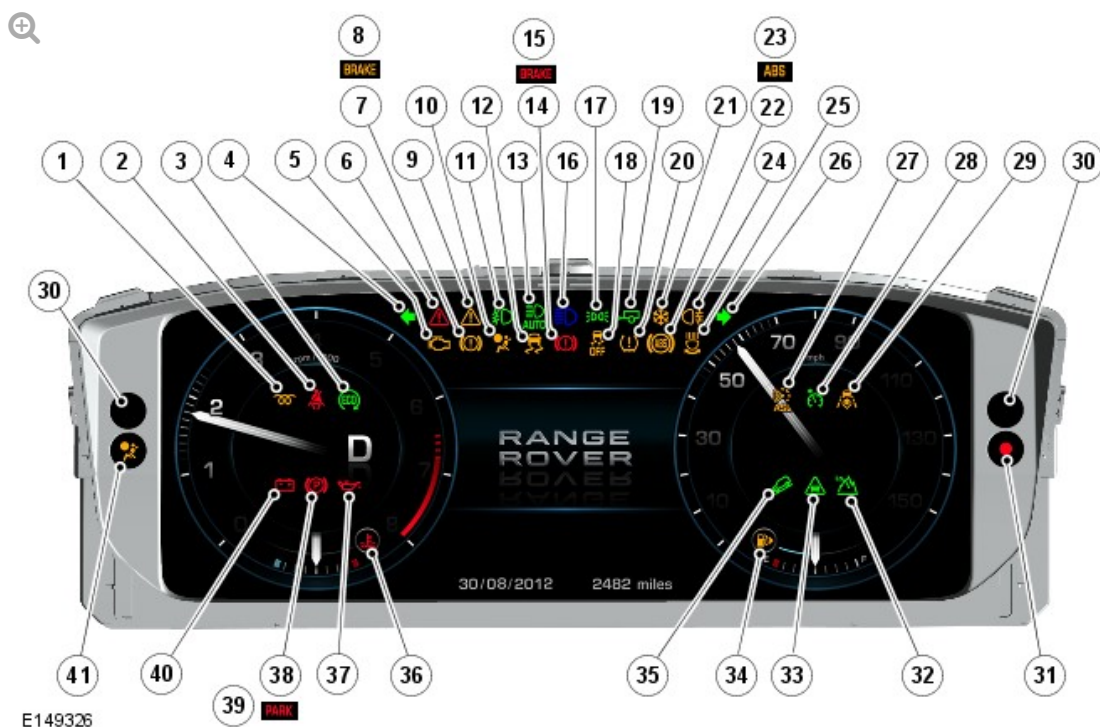
The HUD is designed to work with a specially designed windscreen, which is formed to prevent double images and achieve the best display performance.

When selected the HUD feature projects the following driver information display onto the inside of the windscreen:

- Vehicle speed.
- Traffic Sign Recognition (TSR) and Identified speed limits.
- Gear selection.
- Navigation instructions.
- Cruise control/Automatic Speed Limiter (ASL) active symbol.
- Cruise control/Automatic Speed Limiter (ASL) set speed symbol.
- Follow mode active symbol.

DESCRIPTION

WARNING INDICATORS



ITEM

DESCRIPTION

1	Glow Plug Warning (amber) - Diesel Only
2	Seat Belt Warning (red)
3	Stop/Start (ECO) Active Warning Indicator
4	Left Turn Signal Indicator (green)
5	Check Engine Malfunction Indicator Lamp (MIL) Warning (amber)
6	Critical Warning (red)
7	Brake System Warning (amber) - Rest Of World (ROW)
8	Brake System Warning (amber) - North America Specification (NAS)
9	Warning / Information (amber)
10	Airbag / Supplemental Restraint System (SRS) Warning - Secondary (amber)
11	Front Fog Lamps Active (green)
12	Dynamic Stability Control (DSC) Active Warning (amber)
13	Automatic High Beam Active Warning (amber)
14	Brake System Warning (red) - ROW
15	Brake System Warning (red) - NAS

16	High Beam Warning (blue)
17	Side Lamps On (green)
18	DSC Off Warning (amber)
19	Trailer Warning (green)
20	Tire Pressure Monitoring System (TPMS) Warning Indicator (amber)
21	Frost Warning (amber)
22	Anti-Lock Brake System (ABS) Warning (amber) - ROW
23	ABS Warning (amber) - NAS
24	Rear Fog Lamps Active (amber)
25	Adaptive Front Lighting System (AFS) Warning (amber)
26	Right Turn Signal Indicator
27	Automatic Speed Limiter Active Warning
28	Speed Control Active (green)
29	Adaptive Speed Control Follow Mode Active (amber)
30	Ambient Light Sensor (ref only)
31	Security LED
32	Low Range Selected (green)
33	Forward Alert Active (green)
34	Low Fuel Warning (amber)
35	Hill Descent Control (HDC) Active (green)
36	Oil Temperature Warning (red) (Diesel Only) / Engine Overheat Warning (Petrol Only)
37	Oil Pressure Warning (red) (Diesel Only)
38	Park Brake System Warning (red) - ROW
39	Park Brake System Warning (red) - NAS
40	Charge Indicator (red)
41	Airbag / SRS Warning - Primary (amber)

The warning indicators are mainly located in 3 groups; one in the speedometer display, one in the tachometer display and one group in a central position at the top of the TFT (thin film transistor) screen.

There are 2 airbag / SRS warning indicators used. The primary airbag / SRS warning indicator is located on the left side of the cluster and is a hardwired LED. A secondary airbag / SRS warning indicator is located at the top center of the TFT screen and is only used in case of failure of the primary airbag / SRS warning indicator for legislation requirements.

A hardwired LED is located on the right side of the cluster and is used for the security LED.

START-UP SCREEN



E149327

When the vehicle is locked most of the display goes off. Some vehicle status items remain until the vehicle systems shut down. When the vehicle is unlocked the display slowly illuminates to show the background image and display vehicle starting information to the driver.

Pressing the stop/start button will initiate a full build of the screen display to display the speedometer and tachometer and also the warning indicators. When the engine is running a range of applicable instruments, messages and warning indicators are displayed.

EVENT SCREENS



E149328

A large number of differing permutations of screens are available to cover all possible combinations of the available functions. The above illustration shows the Terrain Response screen, but other screens are available to cover a range of vehicle functions.

Certain vehicle functions will overwrite the existing information displayed. Displays may move or be replaced to allow room for other displays as shown in the Terrain Response display above. For example, the speedometer may move to the right a little to allow room for important vehicle functional to be displayed.

A number of features are also used to convey information to the driver. For example, if speed control is selected, the set speed is displayed as a marker in the speedometer outer ring.

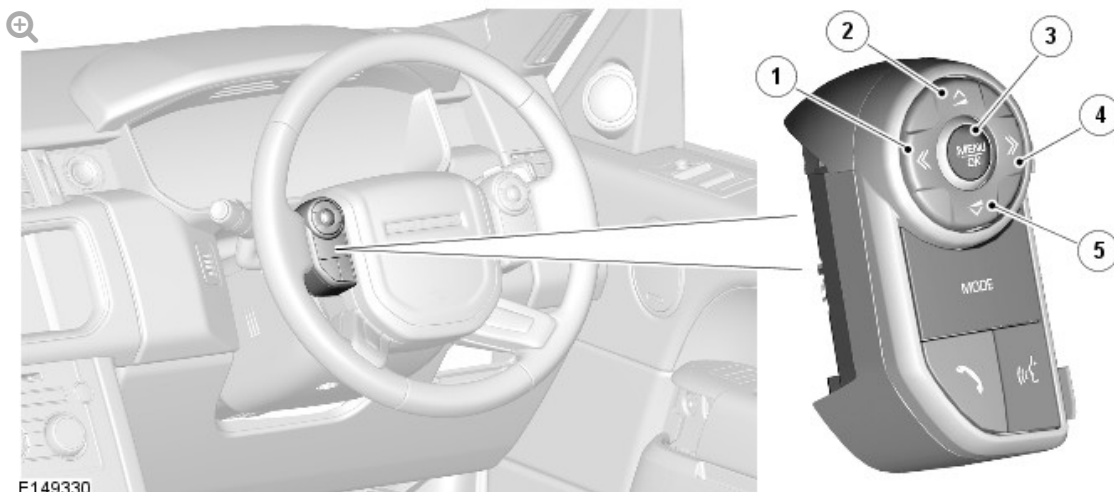
INSTRUMENT CLUSTER (IC) MENU



E149329

The IC menu is only visible when all current messages have been cleared and the MENU/OK switch is pressed. Menu selections are made using the MENU/OK switch to select the required menu option.

MENU/OK Switch



E149330

ITEM	DESCRIPTION
1	Left Selection Switch
2	Up Selection Switch
3	MENU/OK Switch
4	Right Selection Switch
5	Down Selection Switch

The left steering wheel switchpack contains the controls for the IC menu selection. The menu switch comprises a central MENU/OK switch and a four-way switch to allow menu selection UP/DOWN/LEFT/RIGHT. The switchpack is also used to control the audio system.

For additional information, refer to: [Audio System](#) (415-01 Information and Entertainment System, Description and Operation).

Once all current displayed messages are cleared by pressing the MENU/OK switch for each message, the IC menu is displayed with a single press of the MENU/OK switch.

The driver can navigate to the required menu using the UP/DOWN and LEFT/RIGHT switches. When the required menu is highlighted it can be selected using the MENU/OK switch.

When the 'MENU/OK' switch is pressed to select the required menu, the screen will display a sub-menu for that option or activate the selected item. Scroll arrows to the left of the menus show that additional menu items are available that are not visible in the current view. When the arrow is displayed brightly it indicates that additional menu items are available in that direction.

The current displayed menu can be closed by pressing the left switch on the menu switch. If sub-menus are open, pressing and holding the left button on the menu control will close the sub-menus. The menu will close automatically if:

- No selection is made in the main menu within 10 seconds of opening.
- A selection is made in the main menu and no further selection is made within 30 seconds. If a selection is made within the 30 second period, the timer will reset and a further 30 seconds will elapse before the menu automatically closes if no further selections are made.

The current menu can be exited or to move from a sub-menu without making a selection, press the left arrow on the menu control.

The following table shows the available menus and sub-menus and the options available within each menu.

MENU OPTIONS	SUB-MENU	FUNCTION
Main Menu	-	Displays the available main menus.
Driving Features	Show Warnings	Displays all active warning messages which can be scrolled through using the LEFT/RIGHT switches.
	Forward Alert	Enable/Disable forward alert.
	Blind Spot Monitor	Enable/Disable blind spot monitoring system.
	Cruise Control ECO Data	Enable/Disable speed control ECO data view in IC.
	Gearbox Paddles	Allows the selection of the steering wheel paddles to be selected 'Active in Sport Only' mode or 'Active in Drive & Sport' mode.
	High Beam Assist	Enable/Disable the High Beam Assist feature.
		Configure the system for driving on the left or right side of the road.
	TPM Load Setting	Change the monitored tire pressures between 'Heavy Load' and 'Light Load' for the vehicle loading conditions.
Trip Computer	Trip A	Enable/Disable display of Trip A readings.
	Trip B	Enable/Disable display of Trip B readings.
	Trip Auto	Enable/Disable display of Trip Auto readings.
	Units	Select the units (imperial/metric) to be displayed in the message center.
Instrument Display	Display Settings	Off-road View - enable/disable additional suspension information in the touch screen when using the Terrain Response system.

		Torchlight Effect - allows the cluster view to be changed to display all figures on speedometer and tachometer or limited figures in the area of the analogue pointer.
	Speedometer Dial	Display analogue speedometer and select between km/h or mph display units.
	Digital Speedometer	Display digital speedometer and select between km/h and mph display units.
	Language	Allows a number of languages to be selected to display in the IC and touch screen.
	Temperature Units	Allows the temperature to be displayed in Fahrenheit or Celsius.
Vehicle Set-Up		
	Passenger Airbag	Enable/Disable the passenger airbag.
	Alarm Sensors	Temporarily disable the interior space protection and the tilt sensor for the next time the vehicle is locked. The setting is automatically re-enabled when the vehicle is unlocked using the smart key.
	Reverse-dip Mirror	Enable/Disable automatic dipping of the exterior mirrors when reverse gear is selected.
	2-Stage Unlocking	Enable/Disable the single-point entry door unlocking feature.
	Audible Lock Warning	Enable/Disable audible warning when vehicle is locked.
	Windows Global Open	Enable/Disable the window global open feature operated with the smart key.
	Windows Global Close	Enable/Disable the windows global close feature operated with the smart key.
	Winter Wipe Mode	Enable/Disable the winter park position for the front windshield wipers.
	Drive-Away Locking	Disable/Adjust automatic locking of the doors depending on the vehicle speed. The following selections are available; OFF, 8 km/h (5 mph), 16 km/h (10 mph), 32 km/h (20 mph).
	Headlamp Delay	Enable/Disable the headlamp delay feature and configure the amount of time the headlamps will remain on. The following selections are available; OFF, 30 seconds, 60 seconds, 120 seconds, 240 seconds.

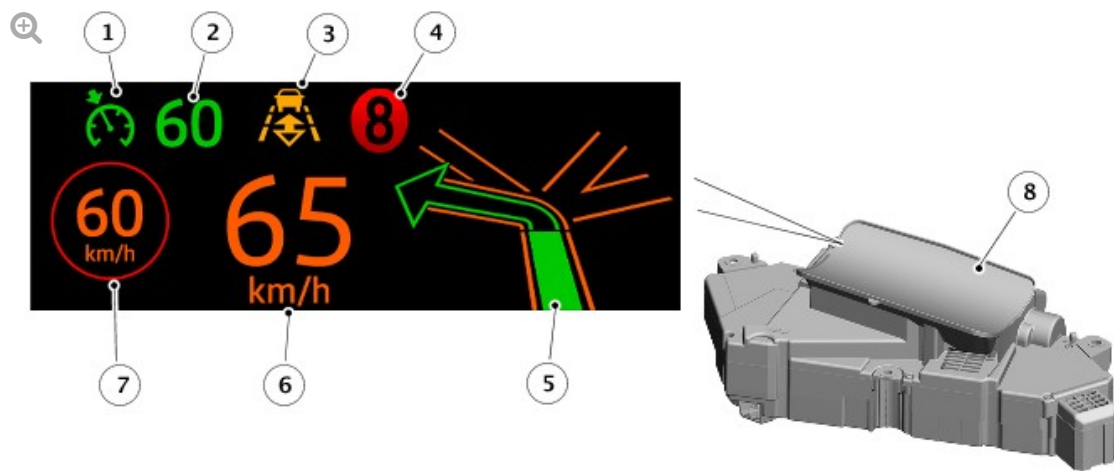
Service Menu	Oil Level	Show the oil level display.
	Last Alarm	Show the cause of the last alarm activation.
	Vehicle VIN	Display the Vehicle Identification Number.
	HBA Sensitivity	Change the sensitivity of the High Beam Assist system. Select between 'Normal Mode' and 'Test Mode'.

CAR CONFIGURATION FILE (CCF)

The Car Configuration File (CCF) contains all relevant data about the specification and market condition of the applicable vehicle, immobilization codes and driver personal settings. This information is retained in the Central Junction Box (CJB) and is read by the IC, enabling the cluster to detect which systems and components are fitted to the vehicle. This information is continuously sent to the IC on the Medium Speed (MS) Controller Area Network (CAN) comfort bus. The IC also backs up the CCF in it's internal memory.

The CCF will also need to be updated using an approved Land Rover diagnostic system if the vehicle is modified in service from its original factory specification. This can include fitting non-standard wheels and/or tyres, optional accessory dealer fit components with an electrical interface, for example park distance control.

HEAD UP DISPLAY (HUD) CONTROL MODULE



E167927

ITEM	DESCRIPTION
1	Speed Control/Adaptive Speed Control Display
2	Speed Control/Adaptive Speed Control Set Speed
3	Follow Mode Warning Indicator
4	Gear Position Display
5	Navigation Turn-By-Turn Display
6	Current Vehicle Speed
7	Traffic Sign Recognition (TSR) Display
8	HUD Control Module

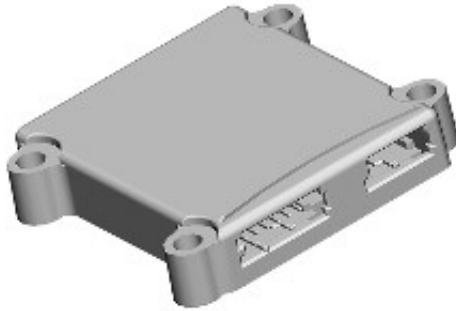
The HUD control module is located inside the instrument panel behind the IC.

The HUD control module receives a permanent battery voltage supply via the Rear Junction Box (RJB). A Medium Speed (MS) Controller Area Network (CAN) comfort bus allows for the HUD control module to communicate with other modules such as HUDCFM and IC.

The HUD uses laser beams within the module to provide the light source, projecting images stored within the laser diode block, via a collimating lens, onto a Liquid Crystal On Silicone (LCOS) surface, with which the beams have a phase relationship. The light is reflected off through a fourier lens and a

series of 'fold' mirrors to create an image on a rotating diffuser (this reduces speckle effects of the laser light). The light is magnified and distorted, in cohesion with the windscreen curvature, and passed across two further freeform curved mirrors, forming the virtual image seen by the driver.

HEAD UP DISPLAY COOLING FAN CONTROL MODULE (HUDCFCM)



E167932

The HUDCFCM is located inside the instrument panel.

The HUDCFCM is powered by the Rear Junction Box (RJB).

Information can be sent or received via the Medium Speed (MS) Controller Area Network (CAN) comfort bus.

The HUDCFCM controls the HUD cooling fan by using a Pulse Width Modulation (PWM) signal.

HEAD UP DISPLAY (HUD) COOLING FAN MOTOR



E167931

The HUD cooling fan is located inside the instrument panel.

The HUD cooling fan is powered by the Rear Junction Box (RJB).

The HUD cooling fan is controlled by a Pulse Width Modulation (PWM) from the HUDCFM.

The HUD cooling fan has a built in sensor to control temperature, so that the motor does not exceed its operating temperature.

OPERATION

INSTRUMENT CLUSTER (IC)

Three processors within the IC manage communications between the vehicle and the IC display and incorporate the operating software required to drive screen re-configurations.

The IC receives a permanent fused supply from the Central Junction Box (CJB).

The IC is connected to other vehicle systems and control modules via the Medium Speed (MS) CAN comfort bus, High Speed (HS) powertrain bus, Local Interconnect Network (LIN) bus and Media Oriented System Transport (MOST) connections. The IC is also connected to the Touch Screen (TS) via a Low-Voltage Differential Signalling (LVDS) connection for video data transmission.

For additional information, refer to: [Audio System](#) (415-01 Information and Entertainment System, Description and Operation).

The majority of information is passed to the IC from other system control modules on the HS and MS CAN buses, LIN bus and MOST. However, some vehicle sensors are hardwired directly to the IC.

The security Light Emitting Diode (LED) is controlled by a hardwired connection from the CJB. A Field Effect Transistor (FET) within the CJB controls the output to the LED.

The Electric Steering Column Lock (ESCL) control module is connected to a hardwired connection to the IC. Security information from other control modules is passed via the network buses and when the conditions are correct the IC instructs the ESCL control module to unlock the steering column.

The left steering wheel switchpack is hardwired to the clockspring, which is connected to the CJB on a LIN bus connection. The CJB then passes the driver selections made on the left steering wheel switchpack to the TS for processing and transmission to the IC and other control modules on the MS CAN comfort bus.

The CJB is connected on the MS CAN body bus and HS CAN powertrain bus to the IC. The IC acts as a gateway for ambient lighting signals from the CJB and converts them to LIN bus signals which are passed from the IC to the interior ambient lighting LED's.

HEAD UP DISPLAY (HUD)

The HUD is controlled from the IC message center menu.

When selected, the menu offers the following options:

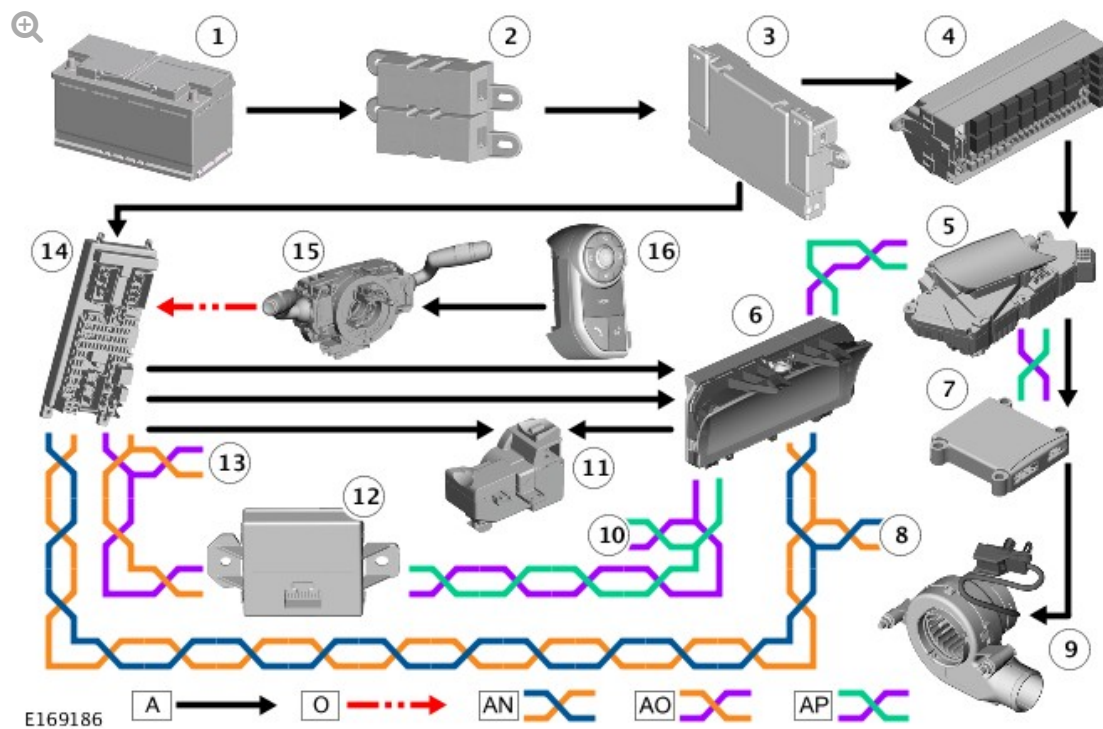
- Selecting the position of the display on the windscreen.
- Display brightness.
- Selecting which information is displayed on the HUD.

Control of the display brightness can be adjusted in three different ways:

- As listed above, manually, from within the IC HUD control message center menu.
- If the vehicle lighting is switched on, the HUD display will adjust simultaneously with the manual adjustment of the IC panel light dimmer switch.
- Automatic brightness, controlled using data from the forward ambient light sensor, housed in the rain sensor.

Prior to selecting the HUD windscreen position the user should select their desired seating position. The steering wheel controls are used to set the horizontal position of the windscreen display. Once set, the information can be saved in the seat memory control system, storing the selected preferences.

CONTROL DIAGRAM



A = HARDWIRED; O = LIN BUS; AN = HS CAN POWERTRAIN BUS; AO = MS CAN BODY BUS; AP = MS CAN COMFORT BUS

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	HUD

6	IC
7	HUDCFM
8	High Speed (HS) CAN Powertrain Bus Connection to Other Vehicle Systems
9	HUD Cooling Fan Motor
10-	Medium Speed (MS) CAN Comfort Bus Connection to Other Vehicle Systems
11	Electric Steering Column Lock (ESCL) Control Module
12	Gateway Module (GWM)
13	Medium Speed (MS) CAN Body Bus Connection to Other Vehicle Systems
14	Central Junction Box (CJB)
15	Clockspring
16	Left Steering Wheel Switchpack